Technological University of the Philippines - Manila
College of Science
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WEKA-BASED DECISION-TREE MODEL FOR USER SUBSCRIPTION PLAN PREDICTION

MEMBERS:
Josef Christopher R. Guban
Christian Dayne R. Menderico
Don Marco G. Montalban

INSTRUCTOR:
Mr. Jonathan Caballero

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ABSTRACT

In today's increasingly competitive streaming landscape, understanding user preferences in subscription plan selection is vital for implementing data-driven strategies. This study examines the impact of key demographic and behavioral attributes—namely age, gender, country, device type, and subscription start month—on the selection of Basic, Standard, or Premium plans on a major streaming platform. Utilizing a dataset of 2,500 anonymized user records, a supervised classification model was developed and validated using the J48 decision-tree algorithm in WEKA. An 80/20 train-test split was applied, and the model's performance was assessed using standard evaluation metrics, including accuracy, Kappa statistic, precision, recall, F-measure, and ROC area.

The model achieved an overall accuracy of 72%, with country identified as the most influential predictor, followed by age and device type. Results revealed interpretable decision rules that highlight how different user profiles correspond to specific subscription preferences. For example, younger smartphone users subscribing later in the year often chose Premium plans, whereas older users on Smart TVs tended toward Standard or Basic tiers. The decision tree effectively captured complex attribute combinations, offering clarity in understanding plan selection behavior.

By transforming raw user data into meaningful patterns, this study underscores the value of interpretable machine learning models in generating actionable insights. These insights can inform subscription plan design and targeted strategy development, illustrating the role of Business Intelligence in bridging user behavior with platform optimization.

CHAPTER I: INTRODUCTION

The Problem and Its Background

In the ever-evolving digital entertainment industry, streaming platforms such as Netflix rely heavily on Business Intelligence (BI) to understand their users and remain competitive. As user behavior becomes increasingly diverse and complex, understanding the factors that influence subscription choices has become a critical area for data-driven decision-making. While many studies have explored content recommendation and user engagement, there remains a gap in analyzing how individual user attributes affect the selection of subscription plans such as Basic, Standard, or Premium.

Consumer choices are often shaped by variables such as age, device preference, and engagement patterns. Identifying these patterns not only improves personalization but also supports marketing strategies, customer retention, and service optimization. Chaurasia and Pal (2019) emphasized that data mining techniques can uncover valuable insights into user decisions, while Dey and Haque (2021) highlighted the importance of predictive analytics in understanding user behavior in subscription-based platforms.

With the growing amount of data available from user interactions, machine learning and classification models offer promising tools for predicting subscription types and analyzing consumer trends. Effective application of these models can lead to enhanced service offerings tailored to different customer segments (Zhang, S., Yao, L., Sun, A., & Tay, Y., 2019). Additionally, García-Gil et al. (2019) noted that preprocessing and intelligent feature selection significantly enhance model performance in large datasets—an important consideration in streaming platforms with millions of active users.

This study explored how user data, such as age, device usage, and viewing behavior, was transformed into meaningful insights using Business Intelligence tools and machine learning models, enabling a deeper understanding of the factors influencing subscription plan selection on a streaming platform like Netflix.

OBJECTIVES

This study aims to derive actionable insights by evaluating user subscription behavior through data analytics.

Specific Objectives:

1. To develop and validate a decision-tree model for accurately predicting users' future subscription plans.
2. To generate data-driven insights from user attributes that influence subscription plan selection.

Scope

This research investigates how a combination of demographic and behavioral attributes shapes subscription-plan choices on an online streaming platform. Leveraging a real-world dataset of 2,500 anonymized user records, the study focuses on five key predictors like country, age, gender, device type, and subscription start month to understand their joint influence on selecting among Basic, Premium, and Standard tiers. The analysis is confined to complete cases for these attributes and employs a decision-tree classifier to extract interpretable rules. Performance is gauged via a percentage split of 80% by 20%, reporting overall accuracy, Kappa statistic, error metrics, and class-specific precision, recall, F-measure, and ROC area. Findings will illuminate which attribute combinations most strongly predict each subscription category, offering actionable insights for targeted marketing and product design within the studied eight-country market.

CHAPTER II: RELATED LITERATURE

Business Intelligence in Subscription Management

Business Intelligence (BI) tools have become central to streaming platforms' ability to segment users, optimize offerings, and drive revenue growth. Nwaimo et al. (2024) highlight that BI software empowers decision-makers to translate large volumes of user data into actionable strategies, from tailoring plan features to forecasting subscription revenues. In the highly fragmented streaming market, platforms leverage dashboards and reporting to track real-time subscription uptake, detect underperforming tiers early, and reallocate marketing spend efficiently (Palani, 2025). This aligns with the growing use of machine learning models, such as decision trees, to uncover patterns in demographic and behavioral data, like age, gender, and device type that influence subscription choices (Garcia & Lee, 2022). Chen and Zhang (2023) further demonstrate that BI-driven user clustering enhances content delivery and subscription plan alignment, leading to higher retention. Park and Sullivan

(2023) also stress the strategic advantage of real-time BI dashboards in responding to shifts in user preferences across regions and subscription tiers. These insights support the analytical approach of this study, which applied classification models to predict user subscription types and guide segmentation strategies across eight countries.

Demographic Drivers of Subscription Choice

A growing body of work examines how user demographics (age, gender, geography, device usage) influence plan selection. Joshi (2024) shows that younger cohorts are more likely to adopt higher‑priced, multi‑device tiers, whereas older segments often remain on basic plans. Some study highlights that demographics and connectivity strongly influence subscription choices. One notable example is Geneva (2023) report in stark international gaps in broadband access and usage, it shows high‑income countries average 257 GB/month per fixed broadband subscription versus only 161 GB in low‑income countries, and 5G mobile coverage is 89% in wealthy countries but nearly 0% in the least developed country. Such disparities imply that consumers in lower‑bandwidth regions (often with fewer smart TVs or 4G devices) may favor basic, lower-cost streaming plans, whereas users in high‑speed markets can more easily choose premium, high‑definition plans. This consistently proves analyses showing that regional broadband and device adoption patterns drive basic and premium subscription preferences.

Predictive Modeling of Subscription Plans

Machine-learning approaches have become essential in forecasting user subscription choices, particularly in dynamic and user-driven platforms like video streaming services. Among various algorithms explored, Akar (2023) compares logistic regression, random forests, and decision trees, concluding that tree-based classifiers strike an ideal balance between predictive accuracy and interpretability, an essential factor for business stakeholders who require transparent decision-making processes. R. Gustavo et al. (2023) further demonstrate how feature-importance scores derived from models can drive personalized marketing campaigns, enabling companies to target users based on their most influential characteristics, such as device type or subscription month. Patil (2024) emphasizes that explainable models like the C4.5 decision tree not only boost transparency but also foster trust in automated systems by providing human-readable rules for strategic marketing.

Furthermore, predictive performance must meet industry reliability thresholds for practical deployment. Hsiao (2023) highlights that an accuracy of at least 70% is considered a benchmark for commercial applications, ensuring the model delivers actionable and generalizable insights. This is particularly important in environments where misclassifying a high-value user could result in substantial revenue loss. Supporting this, Kim, Lee, and Park (2022) applied decision tree models in the context of influencer-driven commerce to predict customer churn, achieving over 80% accuracy. Their work stresses the importance of not only high overall performance, but also strong class-level precision and recall metrics that are critical in capturing the behavior of smaller, more valuable user segments like Premium subscribers in streaming platforms.

The inclusion of behavioral variables such as session length, time of day, and content interaction frequency was also shown to improve model granularity and reduce classification bias. These findings translate directly to subscription-based services, where understanding nuanced user habits can significantly improve tier targeting.

Decision‑Tree Applications in Streaming Analytics

Decision trees have seen wide adoption in streaming analytics because they produce human‑readable rules. Gomez‑Uribe & Hunt (2015) originally applied tree models to viewer engagement, showing how splitting on country, device type, and user tenure yields coherent segments. In the context, J48’s ability to partition on month of subscription start further captures seasonality in plan uptake. Contrary to the notion that deep trees always overfit, recent work shows very large trees can still generalize well with proper validation. For instance, Aouad et al. (2023) report final decision trees with depths 52–78 containing 800–4100 leaves; despite their size these models achieved superior test-set performance (e.g. significant improvements in AUC and lower MSE) when properly cross-validated. In other words, trees with hundreds of leaves can retain strong predictive accuracy if pruned or validated, this substantiates that “trees with 300+ leaves can still generalize well when properly validated. In practice, an 80/20 train-test split is commonly employed to evaluate the performance of decision tree models. This approach ensures that the model is trained on a substantial portion of the data while reserving a separate subset for unbiased testing, thereby providing a reliable assessment of the model's generalization capabilities. Such validation strategies are crucial for developing robust models that perform well on unseen data (Awad & Fraihat, 2023). Emerging research also explores the adaptability of decision trees in streaming data contexts. Manapragada et al. (2020) discuss the use of Hoeffding trees in evolving data streams, highlighting their efficiency in real-time applications. Haug et al. (2022) introduce Dynamic Model Trees for interpretable data stream learning, emphasizing their ability to maintain flexible and robust representations in active data concepts.

Performance Metrics for Subscription Classification

Evaluating predictive models for plan choice requires both aggregate and class‑level measures. Beyond overall accuracy and the Kappa statistic, precision and recall for each tier illuminate where a model may underpredict less‑common plans (e.g., Premium). ROC‑area analysis further quantifies discrimination power across thresholds (Orozco-Arias, 2020). The confusion‑matrix approach provides a granular view of misclassification patterns, crucial for assessing revenue impact when a high‑value user is misassigned to a lower tier. As noted by Sokolova and Lapalme (2009), multi-class performance analysis is essential when class imbalance affects business-critical segments. Powers (2011) emphasizes that relying solely on accuracy may obscure model flaws, especially in uneven class distributions. Similarly, Wu et al. (2021) highlight that ROC and precision-recall curves are more informative in identifying threshold-sensitive errors, particularly in digital consumer modeling. Lastly, Huang and Ling (2005) show that ROC analysis remains robust even under skewed class distributions, making it a reliable tool for evaluating subscription classification systems.

CHAPTER III: METHODOLOGY

Research Design

This study adopts a quantitative, predictive framework in which a supervised decision-tree model (J48 in Weka) is used to classify users into Basic, Premium, or Standard subscription plan based on their profile attributes. An 80 % by 20 % split is applied, 80 % of the data is used for training the model and 20 % for independent testing, ensuring that performance estimates reflect true out-of-sample accuracy. By structuring the work as an experiment in rule induction and validation, researchers systematically uncover patterns in how combinations of country, age, gender, device type, and subscription start month drive plan selection. The use of an interpretable tree structure allows for the derivation of clear, actionable rules that marketing and product teams can leverage. Moreover, framing the analysis as a controlled validation experiment ensures that findings are both reliable and generalizable to new user cohorts.

Data Collection and Preparation

The researchers exported 2,500 anonymized user records from the streaming platform's subscription database, each containing the five predictor variables and the target subscription plan. During preprocessing, duplicate entries and any records with missing values were removed to preserve data integrity and avoid bias in model training (Nasima, 2025). Categorical fields like country, gender, and device type were encoded as nominal attributes, while the subscription start month was formatted as an integer (1–12). Continuous variables such as age were retained to allow the decision-tree algorithm to determine data-driven split points, enhancing the model's ability to capture nonlinear patterns in the data.

Model Training and Validation

The cleaned dataset was subjected to the J48 decision-tree algorithm to construct a hierarchical set of decision rules that map user attributes to subscription outcomes (Razzaq, 2017). To guard against overfitting and to gauge true predictive power, the researchers employed an 80 % by 20 % percentage split, training the model on 80 % of the records and testing it on the remaining 20 %, while preserving class proportions in both subsets. The study recorded overall accuracy, Kappa statistic, mean absolute error, and root mean squared error, as well as precision, recall, F-measure, and ROC area for each subscription category, providing a comprehensive assessment of model performance. Finally, the resulting confusion matrix and per-class ROC curves were analyzed to pinpoint where the model excels and where further refinement is needed.

CHAPTER IV: RESULTS AND DISCUSSION

Decision Tree - Classification Model

The J48 decision tree, trained on 2,499 instances using six attributes, identified country as the most significant predictor of subscription type. Branches further split based on age, gender, device, and start month. In the United States, Smartphone users aged ≤31 who started on or before May were classified as Basic (6 instances). When starting after May and aged ≤30, users were assigned to Premium. Older Smartphone users showed gender-based splits: males aged 33–44 starting mid-year were classified as Standard, while females aged ≤40 using Smart TVs were split between Premium, Basic, or Standard based on finer age and timing (2–9 instances). In Canada, male Smartphone users starting before or in July were mostly classified as Basic or Standard (3–21 instances), while female Tablet users aged ≤41 starting after May were directed to Premium (up to 9 instances). Device and start month were influential across the board.

All United Kingdom users, regardless of demographic or device, were assigned to the Standard plan (195 instances), indicating consistent behavior. In Brazil, most females and males using Tablets or Smart TVs were linked to the Basic plan, especially when starting early in the year (3–23 instances), with occasional variation. Australia showed two Premium-leaning segments: Smartphone users aged ≤28 starting before August (4–6 instances) and older Laptop/Smart TV users starting mid-year (up to 18 instances). In Mexico, users aged >28 were strongly associated with the Standard plan (181 instances), while younger users showed more diversity. Germany's males aged ≤33 were assigned to Basic (19 instances). Older users' classifications varied based on device and start month, with Tablets leaning Basic and others mixed. In France, younger users and early-year Smart TV/Laptop users leaned Premium (up to 26 instances), while older users generally shifted toward Basic, with gender and device also affecting classification.

Table 1: Split Percentage Testing Summary

Correctly Classified Instances	360	72%
Incorrectly Classified Instances	140	28%
Kappa statistic	0.5797	
Mean absolute error	0.2216	
Root mean squared error	0.3882	
Relative absolute error	49.86%	
Root relative squared error	82.34%	
Total Number of Instances	500	

Table 1 summarizes the model's performance on the testing set, including the total number of correct and incorrect classifications, overall accuracy, and the Kappa statistic for agreement between actual and predicted values. It also presents error-based metrics such as mean absolute error, root mean squared error, and their respective relative errors. The total number of evaluated instances is provided, reflecting the dataset size used for validation.

DISCUSSION OF FINDINGS

Table 2. Identifying User Profile Combinations that Predict Specific Subscription Plans

Country	Key Attributes	Likely Subscription
United States	Smartphone users aged ≤31, start_month ≤5	Basic
United States	Smartphone users aged ≤30, start_month >5	Premium
United States	Smart TV users (M) aged 33–44, start_month 5–9	Standard
United States	Smart TV users (F) aged ≤40, start_month ≤10	Premium or Basic
Canada	Male Smartphone users, start_month ≤7	Standard or Basic
Canada	Female Tablet users aged ≤41, start_month >5	Premium
United Kingdom	All users regardless of demographics or device	Standard
Brazil	Most females and males using Smart TVs/Tablets	Basic
Australia	Smartphone users aged ≤28, start_month ≤8	Premium
Australia	Laptop or Smart TV users aged >35, start_month mid-year	Premium
Mexico	Users aged >28	Standard
Germany	Males aged ≤33	Basic
Germany	Older males on various devices	Mixed
France	Younger users and Smart TV/Laptop users, early start_month	Premium

The decision tree reveals that subscription plan selection is strongly influenced by a combination of country, age, device, gender, and start month. In the United States, the model shows a nuanced segmentation: Smartphone users aged 31 and below tend to choose the Basic plan if they subscribe early in the year (start_month ≤ 5), but shift toward Premium when subscribing later (start_month > 5). The preferences of Smart TV users further segment by gender and age. Females aged 40 and below alternate between Basic and Premium, depending on exact age and start month, while males aged 33 to 44 often align with the Standard plan, particularly in the mid-year period. These findings illustrate that subscription behavior in the U.S. varies with time, age, and access device, allowing precise targeting strategies that match subscription plan with user lifecycle stages.

In Canada, the decision rules highlight that male Smartphone users who start their subscription before July typically fall into Standard or Basic, while female Tablet users aged 41 and below tend to choose Premium when starting after May. This underlines the relevance of device type and seasonal timing in Canadian market segmentation. In contrast, the United Kingdom shows a highly uniform trend: regardless of user age, gender, or device, all users fall into the Standard tier. This homogeneity enables simplified campaigns focused solely on promoting the Standard plan’s features without need for personalization.

Other markets exhibit stronger correlations with price sensitivity and device preference. In Brazil, a dominant trend toward the Basic plan is seen among females and Smart TV or Tablet users, particularly with early-year subscriptions, suggesting that affordability and accessibility are key decision factors. In Australia, two distinct Premium segments emerge: younger Smartphone users (aged ≤28) starting before August, and older users using Smart TVs or Laptops, especially mid-year. This indicates dual targeting strategies based on age and device sophistication. In Mexico, the primary determinant is age; users older than 28 are strongly associated with the Standard plan regardless of device, simplifying segmentation efforts. Meanwhile, Germany presents age-divided behavior: younger males (≤33) prefer Basic, but older users show mixed preferences that vary with both device and subscription timing. France demonstrates a stronger lean toward Premium among younger users and Smart TV or Laptop users, especially when subscribing early in the year, while older users transition toward Basic.

Overall, the decision tree structure confirms that user subscription behavior is shaped by complex combinations of demographic and behavioral variables, fulfilling the study's first objective. The insights derived from the model provide a granular understanding of how different segments respond to service offerings, enabling data-driven recommendations for targeted acquisition and retention strategies. These patterns, identified across markets, present actionable frameworks for subscription platforms aiming to optimize user engagement through tailored plan offerings.

Table 3. Predictive Performance of Decision Tree Model

Metric	Basic	Premium	Standard
True Positive Rate (Recall)	0.699	0.724	0.739
Precision	0.711	0.667	0.793
F-Measure	0.705	0.694	0.765
ROC Area	0.816	0.817	0.871

The data supports the decision tree model's predictive performance by showing strong classification results across all subscription plans. The Standard category achieved the highest precision (0.793) and ROC area (0.871), indicating that the model consistently and accurately identifies users in this group. While Basic and Premium had slightly lower precision scores (0.711 and 0.667, respectively), their ROC areas above 0.81 suggest reliable overall performance in distinguishing these classes as well. The F-measure values (0.705 for Basic, 0.694 for Premium, and 0.765 for Standard) reflect a balanced trade-off between precision and recall, with Standard again performing best.

Table 4. Confusion Matrix

basic	premium	standard
123	35	18
32	118	13
18	24	119

Table 4 shows how well the model classified each subscription plan using a confusion matrix. The actual subscription plans are listed in the rows, while the predicted types are in the columns. Correct predictions are found along the diagonal of the matrix, where the actual and predicted classes match. The model correctly identified 123 Basic, 119 Standard, and 118 Premium users. Incorrect predictions appear in the other cells and are fairly evenly spread, indicating that the model performed consistently across the different subscription plans. The confusion matrix further reinforces these findings, highlighting the model's balanced and reliable classification performance across all three subscription plans.

CONCLUSION

This study successfully developed and validated a predictive decision-tree model using the J48 algorithm in Weka to classify users into Basic, Premium, or Standard subscription plans based on key demographic and behavioral attributes. The model achieved an overall accuracy of 72%, with a weighted F-measure of 0.721 and ROC area of 0.834, demonstrating strong predictive power and generalizability across unseen data.

Among the attributes studied, country consistently emerged as the most influential predictor, followed by age, device type, gender, and subscription start month. The classification tree revealed that user preferences are shaped by multi-factor combinations rather than isolated variables. For instance, the same age group may exhibit differing plan preferences based on device usage or start timing, depending on the regional context. This confirms the complexity of user decision-making and highlights the model's ability to reflect real-world behavioral diversity.

The findings affirm that machine learning models, particularly interpretable ones like decision trees, can offer granular insights into subscriber segmentation, enabling streaming platforms to design targeted marketing strategies, adjust pricing tiers, and enhance user acquisition and retention efforts through proactive, data-driven personalization.

RECOMMENDATION

While the model demonstrated solid performance, several areas offer opportunities for enhancement. The dataset was limited to five user attributes, excluding behavioral indicators such as watch time, session frequency, or genre preferences, which may further refine prediction accuracy. Future research should incorporate these variables to capture a broader behavioral context. While the J48 decision tree provided interpretable and effective results, exploring ensemble methods like Random Forests or Gradient Boosted Trees could improve predictive robustness and reduce overfitting. These models may capture complex nonlinear interactions that a single tree might miss, especially in regions with more diverse or ambiguous subscription patterns. It is also recommended to test the model on a different dataset to assess its generalizability and strengthen its integrity across varying data distributions. Furthermore, deploying the model in a real-time setting and evaluating it against live streaming platform data could validate its operational value and help fine-tune the system in evolving markets. This would support the development of adaptive BI tools capable of responding to dynamic consumer behaviors.

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